The Python and Visual Studio Code Configuration Manual

Setting up your Python development environment involves a few key steps, from installing the language itself to configuring your code editor.

Step 1: Install Python

1. **Download Python:** Go to **python.org** and navigate to the download section. Click the large yellow button to download the latest version (e.g., Python 3.13).
2. **Run the Installer:** Open the downloaded file. **Crucially, check the box that says "Add python.exe to PATH"** before clicking "Install Now," as this makes running Python much easier.
3. **Verify the Installation:** Search for your system's **terminal** or command prompt. Type `python --version` and hit enter; if it returns the version number, Python is successfully installed.

Step 2: Install Visual Studio Code (VS Code)

1. **Download VS Code:** Visit **code.visualstudio.com** and click "Download for Windows". If you are on a Mac or Linux, you can find specific versions in the "docs" and "setup" sections of the site.
2. **Install and Open:** Run the installer and open the application once finished.

Step 3: Configure VS Code for Python

1. **Install the Python Extension:** Click the **Extensions icon** on the left sidebar. Search for "Python" and install the official **Python extension by Microsoft** to enable features like smart suggestions and error highlighting.
2. **Set Up a Workspace:** Go to "File" > "Open Folder" and create a dedicated folder for your projects, such as "Python_learning".
3. **Create a Python File:** Click the "New File" icon and name your file with a **.py** extension (for example, `hello.py`). The system will recognize this as a Python file and display a small Python icon.

Step 4: Run Your First Code

1. **Write Code:** Inside your new file, type a simple command like `print("Hi, this is my first Python code")`.
2. **Execute:** Click the **Play icon** in the top right corner of the editor.
3. **Check Output:** Look at the **terminal** panel at the bottom of the screen to see the message you printed.

Step 5: Professional Add-ons (Optional)

- **Custom Shortcuts:** You can set a shortcut (like **Ctrl+R**) to run your code more quickly. Open the Command Palette

(**Shift+Ctrl+P** or **Shift+Cmd+P**), search for "Preferences: Open Keyboard Shortcuts," and search for "run Python file" to bind a key.

- **Formatting:** Install the **autopep8** extension by Microsoft to automatically format your code according to professional standards (PEP 8). Use the Command Palette and search for "Format Document" to clean up your code.
- **Visual Themes:** You can customize the look of your editor by installing themes like **Dracula**, **One Dark Pro**, or **Material Icon Theme** from the extensions menu

Your Ultimate Python Development Setup: A Step-by-Step Guide

1. Download & Install Python

CRUCIAL: Check the "Add python.exe to PATH" box during installation

Ensures the system recognizes Python commands.

2. Verify the Installation

```
$ python --version
> _
```

```
$ python 3.x.x
Python 3.11.4
```

Open your system terminal and type "python –version" to see the installed version.

3. Get Visual Studio Code

Download the appropriate version for Windows, Mac, or Linux from code.visualstudio.com.

Your primary "smart notepad" for writing code.

4. Enable the Python Extension

Open the Extensions marketplace, search for "Python" by Microsoft, and install it for smart suggestions and error highlighting.

5. Initialize Your Project

Create a dedicated folder (e.g., "Python_Learning") and add a new file with the .py extension.

6. Write and Execute "Hello World"

Type the code and click the 'Play' icon in the top right to see the output.

7. Professional Upgrades & Roadmap

Ctrl+Shift+P
Run Python File
Ctrl+R

Set a Professional Shortcut

Map "Run Python File" to a custom keybinding like "Ctrl+R" for faster execution.

Dracula Theme
Material Icon Theme

Aesthetic Enhancements

Install themes for high contrast coloring and distinctive visual icons.

autopep8 Extension

Automate Code Formatting

Install "autopep8" to automatically clean code to PEP 8 styling standards.

Beginner
Intermediate
Advanced

Track Your Progress

Utilize a Notion roadmap to categorize your learning phases and visualize growth.